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Abstract. aaaaaaaaa

1. Introduction

2. The Simplified Model

The general 2D Hindmarsh–Rose-type system is given by:

$$\begin{aligned} \dot{x} &= y - ax^3 + bx^2 + I, \\ \dot{y} &= c - dx^2 - y, \end{aligned} \tag{1}$$

where x represents the membrane potential, y is the recovery variable, and I is the external input current. The parameters a, b, c, d control the intrinsic nonlinearities.

To obtain a mathematically tractable model for global phase-portrait analysis, we fix the parameters as follows:

- $a = 1$
- $c = 1$
- $b = d = k$ (where k is a fixed positive constant, biologically motivated by the standard Hindmarsh–Rose baseline $k = 5$)

The system reduces to:

$$\begin{cases} \dot{x} = y - x^3 + kx^2 + I, \\ \dot{y} = 1 - kx^2 - y, \end{cases} \tag{2}$$

where, I and k are parameters of system.

3. Hopf Bifurcation and Preliminary Analysis of Limit Cycles

We consider the following planar system (2). The aim of this section is to investigate the occurrence of Hopf bifurcation and derive a suitable formulation for the study of periodic solutions.

3.1. **The equilibrium point and Hopf bifurcation.** The equilibrium points of system (2) satisfy

$$y - x^3 + kx^2 + I = 0,$$

and

$$1 - kx^2 - y = 0.$$

Hence system (2) has a unique equilibrium point

$$E_* = (x_*, y_*), \tag{3}$$

where

$$x_* = (1 + I)^{1/3},$$

and

$$y_* = 1 - kx_*^2.$$

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The Jacobian matrix of system (2) is

$$J(x,y)=\begin{pmatrix}-3x^2+2kx & 1\\ -2kx & -1\end{pmatrix}.$$

At the equilibrium point E_* we have

$$J(E_*)=\begin{pmatrix}-3x_*^2+2kx_* & 1\\ -2kx_* & -1\end{pmatrix}.$$

The trace and determinant of this matrix are

$$T=-3x_*^2+2kx_*-1, \tag{4}$$

and

$$D=(-3x_*^2+2kx_*)(-1)+2kx_*. \tag{5}$$

Therefore, $D=3x_*^2$.

The necessary conditions for a Hopf bifurcation are

$$T=0, \qquad D>0.$$

The trace condition gives $-3x_*^2+2kx_*-1=0$, and therefore the Hopf curve is

$$k=k_H=\frac{3x_*^2+1}{2x_*}. \tag{6}$$

And $D=3x_*^2>0$, and $x_*\neq 0$, or equivalently, $I\neq -1$.

So, at the Hopf point, the eigenvalues of $J(E_*)$ are

$$\lambda_{1,2}=\pm i\omega,$$

where

$$\omega=\sqrt{3}|x_*|. \tag{7}$$

It remains to verify the transversality condition. By (4) we obtain $\frac{dT}{dk}=2x_*$. Because $\mathrm{Re}(\lambda)=\frac{T}{2}$, we have

$$\frac{d}{dk}\,\mathrm{Re}(\lambda)=x_*.$$

Therefore,

$$\frac{d}{dk}\,\mathrm{Re}(\lambda)\neq 0$$

whenever $I\neq -1$. Consequently, system (2) undergoes a Hopf bifurcation on the curve (6).

3.2. **The first Lyapunov coefficient.** To determine the type of Hopf bifurcation, we translate the equilibrium point E_* to the origin by introducing

$$x=x_*+u, \qquad y=y_*+v.$$

Then system (2) becomes

$$\begin{cases}\dot u=(-3x_*^2+2kx_*)u+v+(k-3x_*)u^2-u^3,\\ \dot v=-2kx_*u-v-ku^2.\end{cases} \tag{8}$$

At the Hopf point $k=k_H$, using $2k_Hx_*=3x_*^2+1$, the linear part of (8) is

$$A=\begin{pmatrix}1 & 1\\ -(3x_*^2+1) & -1\end{pmatrix}.$$

The nonlinear terms are

$$F_2(u,v)=\begin{pmatrix}(k_H-3x_*)u^2\\ -k_Hu^2\end{pmatrix},$$

and

$$F_3(u,v) = \begin{pmatrix} -u^3 \\ 0 \end{pmatrix}.$$

Using the standard formula for the first Lyapunov coefficient of a planar Hopf bifurcation, after substituting the eigenvectors of A and simplifying the resulting expression, we obtain

$$l_1 = -\frac{3x_*^2+1}{8\sqrt{3}\,x_*^2}. \tag{9}$$

Since $3x_*^2 + 1 > 0$, we conclude that $l_1 < 0$. Hence the Hopf bifurcation is supercritical and a stable limit cycle bifurcates from the equilibrium point when the parameter crosses the Hopf curve. Thus, we have theorem:

Theorem 3.1. *Consider the planar system* (2) *where k and I are real parameters. Let $x_* = (1+I)^{1/3}$, and assume that $I \neq -1$. Then this system has a unique equilibrium point*

$$E_* = \left(x_*, 1 - kx_*^2\right).$$

Moreover, the equilibrium undergoes a supercritical Hopf bifurcation at the parameter value $k = k_H$ (6)*. Hence, for parameter values sufficiently close to and beyond the Hopf curve, system* (2) *possesses a stable limit cycle bifurcating from the equilibrium point E_*.*

3.3. **Local classification of the finite equilibrium.** Since $D > 0$ for all $I > -1$. Therefore, the equilibrium is never a saddle. It is either a node or a focus, and its stability is determined entirely by the sign of the trace T. Set $x_* = (1+I)^{1/3} > 0$. For Hopf value (6) we have

$$T = 2x_*(k - k_H).$$

and

$$T^2 - 4D = 4x_*^2\left[(k-k_H)^2 - 3\right].$$

Hence the node–focus transitions occur at

$$k_{\pm\sqrt{3}} = k_H \pm \sqrt{3}.$$

Since $k_{-\sqrt{3}} = \frac{(\sqrt{3}\,x_*-1)^2}{2x_*} \geq 0$, the local type of E_*, for $k > 0$, is classified as follows:

Parameter range	Type of E_*
$0 < k < k_H - \sqrt{3}$	hyperbolic stable node,
$k = k_{-\sqrt{3}}$	stable improper node,
$k_{-\sqrt{3}} < k < k_H$	hyperbolic stable focus,
$k = k_H$	stable weak focus (Hopf point),
$k_H < k < k_{+\sqrt{3}}$	hyperbolic unstable focus,
$k = k_{+\sqrt{3}}$	unstable improper node,
$k > k_{+\sqrt{3}}$	hyperbolic unstable node.

Therefore, as k increases, the finite equilibrium passes through the sequence

stable node $\longrightarrow$ stable focus $\longrightarrow$ stable weak focus $\longrightarrow$ unstable focus $\longrightarrow$ unstable node.

The values $k = k_{\pm\sqrt{3}}$ mark only node–focus transitions and do not change the stability of the equilibrium; the only local bifurcation occurs at $k = k_H$.

4. Poincaré Compactification and the Dynamics at Infinity

The vector field associated with system (2) is polynomial of degree three. We study its behavior at infinity by Poincaré compactification in the two local charts corresponding to the coordinate axes; see [1].

In the compactification charts, dots denote differentiation with respect to the compactified time τ, rather than the original time t. After each blow-up, primes denote differentiation with respect to the new desingularized time specified there. Dividing a transformed vector field by a common factor m means introducing a time variable satisfying $d\sigma/d\tau = m$. Time orientation is preserved where $m > 0$ and reversed where $m < 0$. These signs must be retained when the resolved field is blown down.

For local chart U_1, use the coordinates

$$x = \frac{1}{v}, \qquad y = \frac{u}{v}.$$

Thus $v = 1/x$ and $u = y/x$. Differentiation gives

$$\frac{du}{dt} = v\left(\frac{dy}{dt} - u\frac{dx}{dt}\right), \qquad \frac{dv}{dt} = -v^2\frac{dx}{dt}.$$

With the positive time rescaling

$$\frac{d\tau}{dt} = v^{-2},$$

substitution yields

$$\begin{aligned} \dot{u} &= u - kv - kuv - uv^2 - u^2v^2 + v^3 - Iuv^3, \\ \dot{v} &= v - kv^2 - uv^3 - Iv^4. \end{aligned} \tag{10}$$

The line at infinity is $v = 0$, and its restricted flow is

$$\dot{u} = u, \qquad \dot{v} = 0.$$

Hence the only singular point on the equator in this chart is $P_1 = (0, 0)$. In the coordinate order (u, v),

$$J_{U_1}(P_1) = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}.$$

Both eigenvalues equal one. Therefore P_1 is a hyperbolic unstable node.

The regions $v > 0$ and $v < 0$ describe the positive and negative horizontal directions, respectively. They represent distinct boundary points of the Poincaré disk, although the signed formulas are the same.

Local chart U_2 is defined by

$$x = \frac{u}{v}, \qquad y = \frac{1}{v}.$$

After the positive time rescaling

$$d\tau = \frac{dt}{v^2},$$

we obtain

$$\begin{aligned} \dot{u} &= -u^3 + ku^3v + ku^2v + uv^2 - uv^3 + v^2 + Iv^3, \\ \dot{v} &= ku^2v^2 + v^3 - v^4. \end{aligned} \tag{11}$$

The infinity line is $v = 0$. On this line,

$$\dot{u} = -u^3, \qquad \dot{v} = 0.$$

Thus the unique singular point on the equator in this chart is

$$P_2 = (0, 0).$$

Its Jacobian satisfies

$$J_{U_2}(P_2) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore P_2 is a linearly zero singular point. Since its linear part gives no information about the local phase portrait, a desingularization by blow-ups is required.

4.1. **Blow-up analysis of the degenerate point in U_2.** In the local chart U_2, the compactified vector field is

$$\begin{aligned} \dot{u} &= -u^3 + ku^3v + ku^2v + uv^2 - uv^3 + v^2 + Iv^3, \\ \dot{v} &= ku^2v^2 + v^3 - v^4. \end{aligned} \tag{12}$$

The line at infinity is $v = 0$, and the restriction of (12) to this line is

$$\dot{u} = -u^3, \qquad \dot{v} = 0.$$

Hence the only singular point on the equator in this chart is $P_2 = (0, 0)$.

Since

$$DX_{U_2}(P_2) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

linearization does not determine its local phase portrait. We therefore resolve the singularity by ordinary directional blow-ups; see [1].

4.1.1. *First directional blow-up.* Let

$$P(u, v) = -u^3 + ku^3v + ku^2v + uv^2 - uv^3 + v^2 + Iv^3$$

and

$$Q(u, v) = ku^2v^2 + v^3 - v^4.$$

The lowest-order nonzero homogeneous terms are

$$P_2(u, v) = v^2, \qquad Q_2(u, v) = 0.$$

Consequently, the characteristic polynomial is

$$\mathcal{F}(u, v) = uQ_2(u, v) - vP_2(u, v) = -v^3.$$

Thus the only characteristic direction is $v = 0$. We perform the directional blow-up

$$v = uz,$$

where $z = v/u$ represents the slope with respect to the u-axis. Since

$$\dot{v} = z\dot{u} + u\dot{z},$$

substitution into the first equation of (12) gives

$$\begin{aligned} \dot{u} &= -u^3 + ku^4z + ku^3z + u^3z^2 - u^4z^3 + u^2z^2 + Iu^3z^3 \\ &= u^2\left(z^2 - u + kuz + uz^2 + Iuz^3 + ku^2z - u^2z^3\right). \end{aligned}$$

Similarly,

$$Q(u, uz) = ku^4z^2 + u^3z^3 - u^4z^4.$$

Therefore

$$u\dot{z} = Q(u, uz) - zP(u, uz),$$

and hence

$$\dot{z} = uz\left(u - z^2 - kuz - Iuz^3\right).$$

Before desingularization, the blown-up system is

$$\begin{aligned} \dot{u} &= u^2\left(z^2 - u + kuz + uz^2 + Iuz^3 + ku^2z - u^2z^3\right), \\ \dot{z} &= uz\left(u - z^2 - kuz - Iuz^3\right). \end{aligned}$$

Both components contain the common factor u. Dividing by this factor gives

$$\begin{aligned} u' &= u\left(z^2 - u + kuz + uz^2 + Iuz^3 + ku^2z - u^2z^3\right), \\ z' &= z\left(u - z^2 - kuz - Iuz^3\right). \end{aligned} \tag{13}$$

Here primes denote differentiation with respect to σ_1, where

$$\frac{d\sigma_1}{d\tau} = u.$$

This divided field reverses the compactified time orientation on the side $u < 0$.

The exceptional divisor is $u = 0$. Restriction of (13) to this divisor gives

$$u' = 0, \qquad z' = -z^3.$$

Hence its only singular point is $(u, z) = (0, 0)$, and

$$DX_1(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus the first directional blow-up does not completely resolve the degeneracy.

The quadratic homogeneous terms of (13) are

$$P_2^{(1)}(u,z) = -u^2, \qquad Q_2^{(1)}(u,z) = uz.$$

The corresponding characteristic polynomial is

$$\mathcal{F}_1(u,z) = uQ_2^{(1)}(u,z) - zP_2^{(1)}(u,z) = 2u^2z.$$

Hence the new characteristic directions are

$$z = 0 \qquad \text{and} \qquad u = 0.$$

We study these directions in complementary blow-up charts.

For completeness, the complementary chart of the first blow-up is

$$u = v\eta.$$

After division by v, with

$$\frac{d\widehat{\sigma}_1}{d\tau} = v,$$

it gives

$$\begin{aligned} \frac{dv}{d\widehat{\sigma}_1} &= v^2 + kv^3\eta^2 - v^3, \\ \frac{d\eta}{d\widehat{\sigma}_1} &= 1 + v(I + k\eta^2 - \eta^3). \end{aligned} \tag{14}$$

Indeed, before division,

$$\frac{d\eta}{d\tau} = \frac{P(v\eta, v) - \eta Q(v\eta, v)}{v}.$$

On the exceptional divisor $v = 0$, the second component equals one. Thus this chart contains no further singular point on that divisor.

4.1.2. *Second blow-up: the direction $z = 0$.* To resolve the direction $z = 0$, introduce

$$z = uw. \tag{15}$$

Since

$$\frac{dz}{d\sigma_1} = w\frac{du}{d\sigma_1} + u\frac{dw}{d\sigma_1},$$

substitution into (13) gives

$$\frac{du}{d\sigma_1} = u^2\left[-1 + kuw + uw^2 + ku^2w + u^2w^2 + Iu^3w^3 - u^4w^3\right],$$

and

$$\frac{dw}{d\sigma_1} = uw\left[2 - 2kuw - 2uw^2 - ku^2w - u^2w^2 - 2Iu^3w^3 + u^4w^3\right].$$

After cancelling the common factor u, set

$$\frac{d\sigma_{2a}}{d\sigma_1} = u.$$

The desingularized system, with primes in σ_{2a}, is

$$\begin{aligned} u' &= u\left[-1 + kuw + uw^2 + ku^2w + u^2w^2 + Iu^3w^3 - u^4w^3\right], \\ w' &= w\left[2 - 2kuw - 2uw^2 - ku^2w - u^2w^2 - 2Iu^3w^3 + u^4w^3\right]. \end{aligned} \tag{16}$$

On the exceptional divisor $u = 0$,

$$u' = 0, \qquad w' = 2w.$$

The only singular point is $(0, 0)$, with Jacobian

$$DX_2(0,0) = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Its eigenvalues are

$$\lambda_1 = -1, \qquad \lambda_2 = 2.$$

Consequently, this point is a hyperbolic saddle, and the characteristic direction $z = 0$ is resolved.

4.1.3. *Second blow-up: the complementary direction* $u = 0$. The direction $u = 0$ is not represented by the previous chart. Introduce the complementary directional blow-up

$$u = zr. \tag{17}$$

Since

$$\frac{du}{d\sigma_1} = r\frac{dz}{d\sigma_1} + z\frac{dr}{d\sigma_1},$$

substitution into (13) produces a common factor z. Divide by this factor and set

$$\frac{d\sigma_{2b}}{d\sigma_1} = z.$$

With primes in σ_{2b}, the resulting system is

$$\begin{aligned} z' &= z\left(r - z - kzr - Iz^3r\right), \\ r' &= r\left(2Iz^3r + kz^2r^2 + 2kzr - z^4r^2 + z^2r + 2z - 2r\right). \end{aligned} \tag{18}$$

The exceptional divisor is $z = 0$. Its restricted flow is

$$z' = 0, \qquad r' = -2r^2.$$

Thus the unique singular point on this divisor is $(z, r) = (0, 0)$, which remains degenerate.

The quadratic homogeneous part of (18) is

$$P_2^{(2)}(z,r) = z(r - z), \qquad Q_2^{(2)}(z,r) = 2r(z - r).$$

Hence

$$\begin{aligned} \mathcal{F}_2(z,r) &= zQ_2^{(2)}(z,r) - rP_2^{(2)}(z,r) \\ &= 2zr(z - r) - rz(r - z) \\ &= 3zr(z - r). \end{aligned}$$

The characteristic directions are therefore

$$z = 0, \qquad r = 0, \qquad r = z.$$

4.1.4. *Third blow-up along the direction $r = z$.* To resolve the direction $r = z$, set

$$r = zs. \tag{19}$$

Since

$$\frac{dr}{d\sigma_{2b}} = s\frac{dz}{d\sigma_{2b}} + z\frac{ds}{d\sigma_{2b}},$$

substitution into (18) and division by the common factor z yield

$$\begin{aligned} z' &= z\left(s - 1 - ksz - Isz^3\right), \\ s' &= s\left(3 - 3s + 3ksz + sz^2 + 3Isz^3 + ks^2z^3 - s^2z^5\right). \end{aligned} \tag{20}$$

Here primes refer to σ, where

$$\frac{d\sigma}{d\sigma_{2b}} = z.$$

The exceptional divisor is $z = 0$, and on it

$$z' = 0, \qquad s' = 3s(1 - s).$$

Consequently, two singular points appear:

$$R_0 = (0, 0), \qquad R_1 = (0, 1).$$

At R_0,

$$DX_3(R_0) = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Thus R_0 is a hyperbolic saddle.

At R_1, the eigenvalues are 0 and -3. Hence R_1 is semi-hyperbolic, and its remaining local dynamics must be determined by a center-manifold reduction.

4.1.5. *Complementary chart of the third blow-up.* The chart $r = zs$ represents the characteristic directions $r = 0$ and $r = z$, but it does not represent the remaining direction $z = 0$. To complete the third blow-up, introduce

$$z = rq.$$

We use the system already given in (18). Since

$$\frac{dz}{d\sigma_{2b}} = q\frac{dr}{d\sigma_{2b}} + r\frac{dq}{d\sigma_{2b}},$$

substitution of $z = rq$ and division by the common factor r give

$$\begin{aligned} q' &= q\left(3 - 3q - 3krq - r^2q^2 - kr^3q^2 - 3Ir^3q^3 + r^5q^4\right), \\ r' &= r\left(-2 + 2q + 2krq + r^2q^2 + kr^3q^2 + 2Ir^3q^3 - r^5q^4\right). \end{aligned} \tag{21}$$

Here primes denote differentiation in $\widehat{\sigma}$, where

$$\frac{d\widehat{\sigma}}{d\sigma_{2b}} = r.$$

The exceptional divisor is $r = 0$. Its restricted flow is

$$q' = 3q(1 - q), \qquad r' = 0.$$

Hence the singular points are

$$Q_0 = (0, 0), \qquad Q_1 = (1, 0).$$

At Q_0,

$$DX(Q_0) = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix},$$

so Q_0 is a hyperbolic saddle.

The point Q_1 does not represent a new singular direction. Indeed, on the overlap of the third-blow-up charts,

$$r = zs, \qquad z = rq,$$

and therefore $sq = 1$. Thus $q = 1$ corresponds to $s = 1$, and $Q_1 = (1, 0)$ represents the same semi-hyperbolic point as $R_1 = (0, 1)$.

Consequently, the two third-blow-up charts cover all three characteristic directions

$$z = 0, \qquad r = 0, \qquad r = z.$$

The direction $z = 0$ gives the hyperbolic saddle Q_0, while the first chart gives the hyperbolic saddle R_0 and the semi-hyperbolic point R_1. We now analyze the latter.

4.1.6. *Center-manifold reduction at $R_1 = (0, 1)$.* Introduce $s = 1 + h$, so that R_1 is translated to $(z, h) = (0, 0)$. System (20) becomes

$$\begin{aligned} z' &= z\left[h - k(1+h)z - I(1+h)z^3\right], \\ h' &= (1+h)\left[-3h + 3k(1+h)z + (1+h)z^2 \right. \\ &\qquad \left. +3I(1+h)z^3 + k(1+h)^2 z^3 - (1+h)^2 z^5\right]. \end{aligned} \tag{22}$$

Its Jacobian at the origin is

$$DX(0,0) = \begin{pmatrix} 0 & 0 \\ 3k & -3 \end{pmatrix}.$$

The eigenvalues are

$$\lambda_1 = 0, \qquad \lambda_2 = -3.$$

For the zero eigenvalue, an eigenvector $(a, b)^T$ satisfies $3ka - 3b = 0$, and hence $b = ka$. Therefore the center eigenspace is

$$E^c = \{(z, h) : h = kz\}.$$

By the center-manifold theorem, a local center manifold, with sufficiently high finite differentiability for the expansions below, can be represented as $h = H(z)$, where

$$H(0) = 0, \qquad H'(0) = k.$$

In particular, $H'(0)$ need not vanish in these coordinates.

Seek an expansion

$$H(z) = a_1 z + a_2 z^2 + a_3 z^3 + O(z^4). \tag{23}$$

The invariance equation is

$$H'(z)z' = h' \qquad \text{on } h = H(z). \tag{24}$$

Substitution of (23) into (24) and comparison of coefficients give, at first order,

$3a_1 - 3k = 0$, so $a_1 = k$.

At second order,

$$4a_1^2 - 7ka_1 + 3a_2 - 1 = 0.$$

Using $a_1 = k$, this reduces to $3a_2 - 3k^2 - 1 = 0$, and therefore

$$a_2 = k^2 + \frac{1}{3}.$$

At third order,

$$-3I - 4a_1^2 k + 9a_1 a_2 - 2a_1 - 8ka_2 + 3a_3 - k = 0.$$

Substitution of the preceding coefficients gives $3a_3 - 3I - 3k^3 - \frac{8k}{3} = 0$. Hence

$$a_3 = I + k^3 + \frac{8k}{9}.$$

The center manifold therefore has the expansion

$$H(z) = kz + \left(k^2 + \frac{1}{3}\right) z^2 + \left(I + k^3 + \frac{8k}{9}\right) z^3 + O(z^4). \tag{25}$$

Equivalently,

$$s = 1 + kz + \left(k^2 + \frac{1}{3}\right) z^2 + \left(I + k^3 + \frac{8k}{9}\right) z^3 + O(z^4).$$

Substitution into the first equation of (22) yields the reduced equation

$$z' = \frac{1}{3}z^3 + \frac{5k}{9}z^4 + O(z^5). \tag{26}$$

Its first nonzero term is $\frac{1}{3}z^3$. Consequently, for sufficiently small nonzero z,

$$z' > 0 \quad \text{if } z > 0, \qquad z' < 0 \quad \text{if } z < 0.$$

The center dynamics is repelling on both sides of R_1. The transverse eigenvalue is -3, so the transverse direction is attracting.

Thus R_1 has saddle-type local behavior for the desingularized field: its center branches are outgoing, and its transverse direction is stable. The uniqueness of the outgoing branches will be established below before the blow-down is completed.

The transformations leading to the chart containing R_1 are

$$v = uz, \qquad u = zr, \qquad r = zs.$$

Combining them gives the exact blow-down relations

$$u = z^2 s, \qquad v = z^3 s.$$

These relations hold throughout the third-blow-up chart.

For trajectories off the exceptional divisor approaching $R_1 = (0, 1)$, we have $s \to 1$. Hence

$$u \sim z^2, \qquad v \sim z^3.$$

Moreover,

$$\frac{v^2}{u^3} = \frac{z^6 s^2}{z^6 s^3} = \frac{1}{s} \longrightarrow 1.$$

Therefore $v^2 \sim u^3$ along such trajectories approaching R_1. This relation follows directly from the blow-down map.

Thus until now we obtained that, the point $P_2 = (0, 0)$ is completely degenerate at the linear level. Directional blow-ups resolve its characteristic directions into hyperbolic saddles and a single semi-hyperbolic point R_1. At R_1, the center-manifold reduction gives

$$z' = \frac{1}{3}z^3 + O(z^4),$$

so the center direction is repelling, while the transverse eigenvalue is negative.

The exact blow-down map near R_1 is $u = z^2 s$, and $v = z^3 s$.

Along trajectories off the exceptional divisor tending to R_1, we have $s \to 1$ and therefore $v^2 \sim u^3$.

The complementary charts cover all characteristic directions, and the first complementary chart contains no additional singular point. The ordinary blow-up atlas and the center-manifold reduction therefore provide a complete local desingularization. A semi-hyperbolic point whose center dynamics has been determined need not be blown up again.

To recover the phase portrait of the original compactified field, we must still restore time orientation, distinguish genuine separatrices from exceptional curves, and collapse the exceptional divisors.

4.2. **Blow-down and the local phase portrait.** We now transfer the resolved dynamics to the compactification charts. In the signed coordinates

$$x = \frac{u}{v}, \qquad y = \frac{1}{v},$$

the half-plane $v > 0$ represents the direction $y \to +\infty$, whereas $v < 0$ represents $y \to -\infty$. These are distinct boundary points of the Poincaré disk, denoted by Y_+ and Y_-, respectively.

The signed chart provides a common algebraic representation of two physical half-charts; their boundary points are not identified. Let τ be defined by $d\tau/dt = v^{-2}$. The compactified equations are

$$\begin{aligned}\frac{du}{d\tau} &= -u^3 + ku^3v + ku^2v + uv^2 - uv^3 + v^2 + Iv^3,\\ \frac{dv}{d\tau} &= ku^2v^2 + v^3 - v^4.\end{aligned}$$

The boundary $v = 0$ is invariant and carries the flow

$$\frac{du}{d\tau} = -u^3.$$

Both boundary branches are directed towards the origin.

The successive substitutions in the chart containing R_1 give

$$u = z^2 s, \qquad v = z^3 s.$$

Write its local invariant center curve as $s = S(z)$, where

$$S(z) = 1 + kz + \left(k^2 + \frac{1}{3}\right) z^2 + \left(I + k^3 + \frac{8k}{9}\right) z^3 + O(z^4).$$

Along this curve, in the final desingularized time,

$$\frac{dz}{d\sigma} = \frac{1}{3} z^3 + \frac{5k}{9} z^4 + O(z^5). \tag{27}$$

The successive divisions along this sequence are by u, z, and z. Consequently,

$$\frac{d\sigma}{d\tau} = uz^2 = z^4 s.$$

Near R_1, we have $s > 0$, and therefore

$$\frac{d\sigma}{d\tau} > 0 \qquad \text{for } z \neq 0.$$

Since $d\tau/dt > 0$, time orientation is preserved near both center branches. Equation (27) shows that their forward orientation is away from infinity.

For $\varepsilon \in \{-1, 1\}$, denote by Γ_ε the branch parametrized by

$$(u, v) = \big(z^2 S(z), z^3 S(z)\big), \qquad 0 < \varepsilon z \ll 1. \tag{28}$$

Since $S(z) > 0$ near zero, Γ_+ lies in $v > 0$, whereas Γ_- lies in $v < 0$. Each physical vertical point thus has one such outgoing interior branch. Moreover,

$$\frac{v^2}{u^3} = \frac{1}{S(z)} \longrightarrow 1, \qquad v \sim \varepsilon u^{3/2} \quad \text{on } \Gamma_\varepsilon.$$

The corresponding asymptotics in the original phase plane follow from

$$x = \frac{1}{z}, \qquad y = \frac{1}{z^3 S(z)}.$$

Indeed,

$$\frac{1}{S(z)} = 1 - kz - \frac{1}{3} z^2 - \left(I + \frac{2k}{9}\right) z^3 + O(z^4).$$

Therefore

$$y = x^3 - kx^2 - \frac{x}{3} - I - \frac{2k}{9} + O(|x|^{-1}), \qquad x \to \varepsilon\infty. \tag{29}$$

In particular, $y/x^3 \to 1$. Substitution into the original first equation gives

$$\frac{dx}{dt} = -\frac{x}{3} - \frac{2k}{9} + O(|x|^{-1}),$$

confirming that the forward flow on either branch is directed from infinity towards the finite plane.

Uniqueness of the outgoing interior branch. Let $f(z,h)$ and $g(z,h)$ denote the right-hand sides of (22), and use the invariant graph $h = H(z)$ constructed above. Set $\eta = h - H(z)$. Its exact differential equation is

$$\frac{d\eta}{d\sigma} = g(z, H(z)+\eta) - H'(z) f(z, H(z)+\eta).$$

The invariance equation makes the right-hand side zero at $\eta = 0$. The fundamental theorem of calculus therefore gives $\frac{d\eta}{d\sigma} = a(z,\eta)\eta$, where

$$a(z,\eta) = \int_0^1 \left[g_h(z, H(z)+\theta\eta) - H'(z) f_h(z, H(z)+\theta\eta) \right] d\theta.$$

Since $g_h(0,0) = -3$, and $f_h(0,0) = 0$, we have $a(0,0) = -3$. After reducing the neighborhood, $a(z,\eta) \le -2$.

Suppose an orbit tends to R_1 as $\sigma \to -\infty$. If $\eta(\sigma_0) \neq 0$ at a sufficiently negative time, then, for $\sigma < \sigma_0$,

$$\begin{aligned} |\eta(\sigma)| &= |\eta(\sigma_0)| \exp\left(\int_{\sigma_0}^{\sigma} a(z(\zeta), \eta(\zeta))\, d\zeta \right) \\ &\ge |\eta(\sigma_0)| e^{2(\sigma_0 - \sigma)}. \end{aligned}$$

This contradicts convergence to R_1. Hence every such orbit lies on $\eta = 0$, that is, on the center graph.

On this graph, (26) has exactly one local orbit on each interval $z > 0$ and $z < 0$. Both approach zero in backward time, because their speed is $z^3/3 + O(z^4)$ and the corresponding time integral diverges at zero.

Thus there is exactly one outgoing branch on each side of R_1. After blow-down, this gives one outgoing interior branch at each physical vertical point. The argument establishes uniqueness of actual invariant branches, rather than only uniqueness of their formal expansions.

Exceptional divisors and genuine separatrices. The final ordinary charts have the following blow-down maps and cumulative time factors:

Chart	Blow-down map	Cumulative time factor
(u,w)	$(u,v) = (u, u^2 w)$	$d\sigma_{2a}/d\tau = u^2$
(z,s)	$(u,v) = (z^2 s, z^3 s)$	$d\sigma/d\tau = z^4 s$
(q,r)	$(u,v) = (r^2 q, r^3 q^2)$	$d\widehat{\sigma}/d\tau = r^4 q^2$

(30)

These formulas distinguish invariant curves that collapse to the origin from curves that remain nontrivial after blow-down.

In the (u,w) chart, denote the saddle of (16) by $A_0 = (0,0)$. The axis $u = 0$ is exceptional, whereas $w = 0$, $u \neq 0$, maps to the original boundary $v = 0$. The eigenvalue along $w = 0$ is -1, and the cumulative time factor is positive. The two signs of u therefore represent the two incoming boundary branches at a fixed physical vertical point.

At $R_0 = (0,0)$ in the (z,s) chart, both invariant axes $z = 0$ and $s = 0$ collapse to $(u,v) = (0,0)$. Likewise, at $Q_0 = (0,0)$ in the (q,r) chart, both invariant axes $q = 0$ and $r = 0$ are exceptional. Because these points are hyperbolic saddles, their local separatrices

are precisely these axes. They create no additional interior separatrix of the original compactified field.

At R_1, the invariant divisor $z = 0$ is the stable separatrix. The center branches satisfy $z \neq 0$ and $s = S(z) > 0$; they are nonexceptional and blow down to Γ_+ and Γ_-. Thus the only candidate interior separatrix in each physical half-chart is the unique branch already constructed.

Time orientation along the divisor connections. Fix a physical half-chart $\varepsilon v > 0$. In the (u, w) chart, this means $\varepsilon w > 0$. On $u = 0$, the divided flow is $w' = 2w$. The cumulative time factor is $u^2 > 0$ off the divisor. Hence the divisor arrows, interpreted from the adjacent physical region, point away from A_0 towards increasing $|w|$.

On the overlap with the second complementary chart, $r = \frac{1}{w}$ and $z = uw$.

This flow therefore approaches $r = 0$ along the old second divisor. In the third complementary chart, that divisor is $q = 0$, where $r' = -2r$. It consequently enters Q_0.

On the new divisor $r = 0$ in the (q, r) chart, $q' = 3q(1 - q)$.

The cumulative factor r^4q^2 is positive off the divisor. For $0 < q < 1$, the arrows point from Q_0 towards Q_1, which represents R_1. For $q < 0$, the arrows point towards $q \to -\infty$. Under $s = 1/q$, this is the side $s \to 0^-$ of R_0.

The (z, s) chart requires a sign correction because its cumulative time factor z^4s has the sign of s. On $z = 0$, the divided equation is $s' = 3s(1 - s)$. The physical arrows are therefore reversed for $s < 0$. They point towards $s = 0$ on the negative-s side and towards $s = 1$ for $0 < s < 1$. For $s > 1$, they also point towards $s = 1$.

On the old first divisor $s = 0$, the divided equation is $z' = -z$. From the adjacent region $s < 0$, physical time reverses these arrows, so they point away from R_0. From the adjacent region $s > 0$, they point towards R_0.

The first complementary chart connects these two incidences of the old first divisor. In that chart, $u = v\eta$ and the divided flow on $v = 0$ is $\eta' = 1$.

Since $d\widehat{\sigma}_1/d\tau = v$, its physical arrows have the sign of ε. They run from

$$\eta = -\varepsilon\infty \quad \text{to} \quad \eta = \varepsilon\infty$$

through regular finite values of η. Because $\eta = u/v$, this connects the negative-u side to the positive-u side. It joins the outgoing incidence of R_0 adjacent to $s < 0$ to the incoming incidence adjacent to $s > 0$. From there, the divisor flow proceeds towards R_1.

Consequently, the divisor chain adjacent to the positive-u boundary branch reaches R_1 through $0 < q < 1$. The chain adjacent to the negative-u boundary branch first passes through $q < 0$, the negative-s incidence of R_0, and the regular first divisor; it then reaches R_1 through $0 < s < 1$.

Different incidences here distinguish sides of signed blow-up charts, not additional equilibria of the original compactification. These two chains exhaust the ordinary atlas and contain no directed cycle.

Reconstruction of the two sectors. Each of the two chains starts at an incoming boundary branch and ends at the stable divisor direction of R_1. Its intermediate singularities are hyperbolic saddles whose separatrices lie on exceptional divisors. At each regular point of a chain, nearby trajectories pass through a flow box. The relevant saddle sectors and regular flow boxes therefore join to form a passage from a neighborhood of the incoming boundary branch to a neighborhood of the outgoing center branch.

Collapsing the exceptional divisors turns each passage into one hyperbolic sector at Y_ε, bounded by an incoming boundary separatrix and Γ_ε. This is the usual sector reconstruction under blow-down.

The complete chart coverage, absence of further nonexceptional separatrices, and directed connections above exclude additional sectors. Thus there are exactly two hyperbolic sectors on the disk side of each Y_ε, separated by its unique outgoing interior branch.

Here a hyperbolic sector means a saddle sector; the original equilibrium itself is not hyperbolic, since its Jacobian is zero. Each vertical point attracts along the boundary and has one outgoing interior separatrix. Attraction along the boundary does not make that point an attractor for finite trajectories.

4.3. **The flow on the Poincaré boundary.** Represent the compactified plane by the closed upper hemisphere

$$\mathbb{S}^2_+ = \left\{(X, Y, Z) : X^2 + Y^2 + Z^2 = 1,\ Z \geq 0\right\},$$

whose equator is identified with the boundary of the Poincaré disk. Write

$$X_\pm = (\pm 1, 0, 0), \qquad Y_\pm = (0, \pm 1, 0).$$

Although the points in each pair are antipodal, they remain distinct in this compactification of the plane.

The homogeneous terms of degree three in the original vector field are

$$P_3(x, y) = -x^3, \qquad Q_3(x, y) = 0.$$

With $(X, Y) = (\cos\theta, \sin\theta)$ on the equator, a positive normalization of compactified time gives

$$\begin{aligned} \theta' &= \cos\theta\, Q_3(\cos\theta, \sin\theta) - \sin\theta\, P_3(\cos\theta, \sin\theta) \\ &= \sin\theta \cos^3\theta. \end{aligned} \tag{31}$$

Its only zeros are $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, corresponding to X_+, Y_+, X_-, and Y_-. The signs of the right-hand side give

Boundary arc	Forward orientation
$0 < \theta < \pi/2$	$X_+ \longrightarrow Y_+$
$\pi/2 < \theta < \pi$	$X_- \longrightarrow Y_+$
$\pi < \theta < 3\pi/2$	$X_- \longrightarrow Y_-$
$3\pi/2 < \theta < 2\pi$	$X_+ \longrightarrow Y_-$

(32)

Each open arc is a heteroclinic orbit of the boundary flow.

The transverse behavior at the horizontal points can also be read from the physical charts

$$x = \frac{\varepsilon}{a}, \qquad y = \frac{b}{a}, \qquad a \geq 0, \qquad \varepsilon \in \{-1, 1\}.$$

After the positive time change $d\tau/dt = a^{-2}$, the linearization in the coordinate order (a, b) is

$$\begin{pmatrix} 1 & 0 \\ -k & 1 \end{pmatrix}.$$

Thus both X_+ and X_- are hyperbolic source nodes.

At the vertical points, the blow-down analysis supplies the transverse information absent from the scalar boundary equation: each has two incoming boundary branches, one outgoing interior separatrix, and two hyperbolic sectors on the disk side.

Thus for classification of the dynamics at infinity we can prove the following theorem:

Theorem 4.1. *Consider system* (2)*. Its Poincaré compactification has exactly four equilibria on the boundary of the compactified plane, namely $X_\pm$ and $Y_\pm$. The points $X_\pm$ are hyperbolic source nodes. Each Y_ε, $\varepsilon \in \{-1, 1\}$, is a degenerate boundary saddle with two hyperbolic sectors on the disk side, two incoming boundary separatrices, and a unique local outgoing interior separatrix Γ_ε.*

In the signed coordinates $u = x/y$, $v = 1/y$, this interior separatrix satisfies

$$v \sim \varepsilon u^{3/2}, \qquad u \to 0^+.$$

Equivalently, in the original variables,

$$y = x^3 - kx^2 - \frac{x}{3} - I - \frac{2k}{9} + O(|x|^{-1}), \qquad x \to \varepsilon\infty.$$

Its local forward orientation is from infinity into the finite plane.

The four open boundary arcs are heteroclinic orbits directed from the horizontal points to the adjacent vertical points:

$$X_+ \longrightarrow Y_+, \qquad X_- \longrightarrow Y_+, \qquad X_- \longrightarrow Y_-, \qquad X_+ \longrightarrow Y_-.$$

There are no other boundary equilibria or periodic boundary orbits. The boundary connections and the local sector configuration at infinity are independent of k and I.

Proof. Equation (31) identifies all boundary equilibria and gives the four connections in (32). The horizontal chart linearizations have two positive eigenvalues, proving that $X_\pm$ are hyperbolic source nodes.

For each vertical direction, the ordinary blow-up charts resolve the degeneracy into hyperbolic saddles and the semi-hyperbolic point R_1. At R_1, the center equation is

$$z' = \frac{1}{3}z^3 + O(z^4),$$

and the transverse eigenvalue is -3. The exact transverse equation in Subsection 4.2 proves uniqueness of each outgoing center branch.

The remaining resolved saddle separatrices either lie on exceptional divisors or map to incoming boundary branches. The ordinary-chart connections, with their physical time orientations restored, give exactly two hyperbolic sectors on the disk side of each vertical point.

Near R_1, the cumulative time factor $z^4 s$ is positive off the divisor. Hence blow-down preserves the outgoing orientation of the interior branch. Its asymptotic expressions follow from (28)–(29).

Finally, the angular field is nonzero on each open boundary arc and vanishes at its endpoints. Consequently, no periodic orbit lies on the boundary. Neither the angular signs nor the eigenvalue signs and leading cubic coefficient in the local resolutions depend on the parameters. This proves the stated parameter independence of the configuration at infinity. □

The theorem classifies the boundary flow and its adjacent local sectors. The global continuations of the interior separatrices, and the number and stability of periodic orbits in the finite plane, require a separate analysis.

5. Global existence and the number of limit cycles

The affine coordinates

$$p = x - a, \qquad q = x + y - (a + 1 - ka^2)$$

place the equilibrium at the origin and give the Liénard system

$$\dot{p} = q - F_k(p), \qquad \dot{q} = -g(p),$$

where

$$F_k(p) = p\big[p^2 + (3a - k)p + 3a^2 - 2ka + 1\big],$$
$$g(p) = (p + a)^3 - a^3 = p(p^2 + 3ap + 3a^2).$$

Thus $pg(p) > 0$ for $p \neq 0$. The associated potential is

$$G(p) = \int_0^p g(s)\, ds = \frac{1}{4}p^4 + ap^3 + \frac{3}{2}a^2p^2 = \frac{p^2}{4}\big[(p + 2a)^2 + 2a^2\big]. \tag{33}$$

It is positive away from the origin and tends to infinity as $p \to \pm\infty$. The variable q is the Liénard coordinate $\dot{p} + F_k(p)$; it is not the velocity $\dot{p}$.

The main result determines the existence of periodic orbits for every $I > -1$ and gives their exact global number for every $k > 0$ under the explicit sufficient condition

$$a \geq a_0 := \sqrt{\frac{3}{17}}, \qquad \text{equivalently} \qquad I \geq I_0 := \left(\frac{3}{17}\right)^{3/2} - 1 \approx -0.925868. \tag{34}$$

In particular, this condition includes all $I \geq 0$. The proof concerns every periodic orbit in $\mathbb{R}^2$; there is no restriction to small amplitudes or to a neighborhood of the equilibrium. For the remaining input values, we retain the previously established uniqueness regions. Define

$$K(a) = \max\left\{a + \frac{1}{a},\ 3a + 3\sqrt{3}\right\}. \tag{35}$$

Since $K(a) \geq a + 1/a$ and $K(a) > 2a$, we have $K(a) > [(a + 1/a) + 2a]/2 = k_H$.

Theorem 5.1 (Global dynamics and the number of limit cycles)**.** *Let $I > -1$ and $k > 0$, and retain the definitions of a, E_*, k_H, I_0, and $K(a)$ introduced above. Then the following statements hold.*

(1) Every solution of (2) exists for all $t \geq 0$ and is bounded. Moreover, a compact positively invariant set absorbs every forward trajectory.

(2) If $0 < k \leq k_H$, then E_* is globally asymptotically stable. Consequently, there are no periodic orbits.

(3) If $k > k_H$, then E_* is a hyperbolic source, and the system has a finite, nonzero number of limit cycles. At least one of these cycles is orbitally asymptotically stable, and every non-equilibrium solution approaches a limit cycle as $t \to \infty$.

(4) If $I \geq I_0$, then for every $k > k_H$ the system has exactly one limit cycle in $\mathbb{R}^2$.

If $-1 < I < I_0$, then there exists $\delta(a) > 0$ such that the system has exactly one limit cycle in $\mathbb{R}^2$ whenever

$$k_H < k < k_H + \delta(a) \qquad \text{or} \qquad k > K(a).$$

In all these cases, the unique limit cycle is hyperbolic and orbitally asymptotically stable. Its basin of attraction is $\mathbb{R}^2 \setminus \{E_*\}$.

Proof. We first establish a bound valid for every initial condition and forward completeness. So, we Choose

$$c = \frac{1}{2(1 + k^2)}, \qquad W(x, y) = x^2 + cy^2.$$

Along a solution,

$$\frac{1}{2}\dot{W} = xy - x^4 + kx^3 + Ix + cy - ckx^2y - cy^2.$$

Young's inequality yields

$$|xy| \leq \frac{c}{8}y^2 + \frac{2}{c}x^2, \qquad c|y| \leq \frac{c}{8}y^2 + 2c,$$

and, since $ck^2 \leq 1/2$,

$$ckx^2|y| \leq \frac{1}{4}x^4 + c^2k^2y^2 \leq \frac{1}{4}x^4 + \frac{c}{2}y^2.$$

Consequently,

$$\frac{1}{2}\dot{W} \leq -\frac{3}{4}x^4 + k|x|^3 + \frac{2}{c}x^2 + |I||x| + 2c - \frac{c}{4}y^2.$$

For example, the explicit constant

$$C = \frac{729}{4}k^4 + \frac{12}{c^2} + \frac{3^{4/3}}{4}|I|^{4/3} + 2c$$

satisfies

$$k|x|^3 + \frac{2}{c}x^2 + |I||x| + 2c \leq \frac{1}{4}x^4 + C.$$

Using also $x^4 \geq x^2/2 - 1/16$, we obtain

$$\dot{W} \leq -\frac{1}{2}W + C_1, \qquad C_1 = 2C + \frac{1}{16}. \tag{36}$$

Integration gives

$$W(t) \leq e^{-t/2}W(0) + 2C_1(1 - e^{-t/2}). \tag{37}$$

This estimate holds throughout the maximal forward existence interval and prevents finite-time escape. The usual continuation theorem therefore gives a global forward solution; see [4].

For every $R > 2C_1$, the ellipse

$$\mathcal{K}_R = \{(x, y) : W(x, y) \leq R\}$$

is positively invariant, because $\dot{W} < 0$ on its boundary. Equation (37) shows that every trajectory enters it in finite time. Every periodic orbit is contained in $\{W \leq 2C_1\}$: at a maximum of W along a periodic orbit, (36) implies $0 \leq -W/2 + C_1$.

These estimates can be made uniform when k varies over a compact interval. Indeed, replace k by an upper bound κ when choosing c and C. This uniform version will be used below to exclude distant cycles near Hopf.

Now, we use the equal-potential comparison criterion of Sugie and Hara, recalled in [2]. For a Liénard system with $pg(p) > 0$ and $G(\pm\infty) = \infty$, let $p_L(h) < 0 < p_R(h)$ be the unique points satisfying $G(p_L(h)) = G(p_R(h)) = h$, where $h > 0$. If

$$F_k(p_R(h)) \neq F_k(p_L(h)) \qquad \text{for every } h > 0,$$

then no periodic orbit exists. This criterion compares the two sides of the equilibrium at the same potential value and does not require symmetry.

For the present system, put $x_L = a + p_L(h)$, $x_R = a + p_R(h)$, and $S = x_L + x_R$. In the original x-coordinate the potential is

$$\widehat{G}(x) = G(x - a) = \frac{1}{4}x^4 - a^3x + \frac{3}{4}a^4.$$

Subtracting the equal potential values gives $S(x_L^2 + x_R^2) = 4a^3$. In particular, $S > 0$. Since $x_L \neq x_R$, $x_L^2 + x_R^2 > S^2/2$, and therefore $0 < S < 2a$. Writing $\Phi_k(x) = x^3 - kx^2 + x$, we have $F_k(p) = \Phi_k(p + a) - \Phi_k(a)$. Hence

$$\begin{aligned} F_k(p_R(h)) - F_k(p_L(h)) &= (x_R - x_L)\big[x_R^2 + x_R x_L + x_L^2 - kS + 1\big] \\ &= (x_R - x_L)S\big[J_a(S) - k\big], \end{aligned} \tag{38}$$

where

$$J_a(S) = \frac{S}{2} + \frac{1}{S} + \frac{2a^3}{S^2}.$$

For $0 < S \leq 2a$,

$$J_a'(S) = \frac{1}{2} - \frac{1}{S^2} - \frac{4a^3}{S^3} < 0, \qquad J_a(2a) = k_H.$$

Thus $J_a(S) > k_H \geq k$ whenever $0 < k \leq k_H$. All factors on the right of (38) are positive, so the comparison criterion excludes periodic orbits throughout this entire parameter range.

The equilibrium is locally asymptotically stable for $k < k_H$. At $k = k_H$, the negative first Lyapunov coefficient established in the Hopf calculation makes it an attracting weak focus, so local asymptotic stability also holds at the endpoint; see [5]. By boundedness, every forward trajectory has a nonempty compact omega-limit set. If this set contained no equilibrium, the Poincaré–Bendixson theorem would make it a periodic orbit. Since periodic orbits have just been excluded and the equilibrium is unique, the omega-limit set contains E_*. The trajectory consequently enters its local basin of attraction and converges to E_*. Local Lyapunov stability and convergence from every initial condition prove global asymptotic stability.

Now, for global periodic behavior above Hopf, we fix $k > k_H$. Let $\xi = (x-a, y-1+ka^2)^T$ and $A = DX(E_*)$. Since A has eigenvalues with positive real parts, the matrix

$$M = \int_0^\infty e^{-A^T t} e^{-At}\, dt$$

is positive definite and satisfies $A^T M + MA = \mathrm{Id}_2$. Therefore, for $L(\xi) = \xi^T M \xi$,

$$\dot{L} = \|\xi\|^2 + O(\|\xi\|^3) > 0$$

in a sufficiently small punctured neighborhood of the equilibrium. Choose $\varepsilon > 0$ small enough that the boundary of $\mathcal{U} = \{L \le \varepsilon\}$ is crossed strictly outward. A non-equilibrium trajectory inside $\mathcal{U}$ leaves it in finite time: after decreasing ε if necessary, $\dot{L} \ge c_0 L$ there for some $c_0 > 0$. Once outside, it cannot return.

Take an absorbing ellipse large enough that $\mathcal{U} \subset \operatorname{int} \mathcal{K}_R$. Every non-equilibrium trajectory eventually lies in the compact positively invariant annulus

$$\mathcal{A} = \mathcal{K}_R \setminus \operatorname{int} \mathcal{U}.$$

There is no equilibrium in $\mathcal{A}$. The Poincaré–Bendixson theorem consequently shows that the omega-limit set of every such trajectory is a periodic orbit; see [1].

We next show that these periodic orbits are isolated and finite in number. Introduce the velocity $v = \dot{x}$. The equivalent system is

$$\dot{x} = v, \qquad \dot{v} = a^3 - x^3 - (3x^2 - 2kx + 1)v. \tag{39}$$

Consider the transversal $\Sigma = \{x = a,\ v > 0\}$. For an initial velocity $v_0 > 0$, define $P(v_0)$ as the velocity at its first return to Σ.

This return exists for every $v_0 > 0$. On $x > a$, a zero of v has $\dot{v} = a^3 - x^3 < 0$, so a rightward motion can turn only toward the left. On $x < a$, a zero of v has $\dot{v} > 0$, so a leftward motion can turn only toward the right. Boundedness rules out escape, and the source cannot be approached by a non-equilibrium trajectory. These facts force successive crossings of $x = a$ in the negative and positive directions. More explicitly, a bounded trajectory that moved monotonically in x forever would approach $(\ell, 0)$; the second equation in (39) would then force $\ell = a$, contradicting the source property.

The polynomial vector field and the transverse crossings make P real analytic on $(0, \infty)$. In fact, if $\tau(v_0)$ denotes the return time, the variational equation gives

$$P'(v_0) = \frac{v_0}{P(v_0)} \exp\left[- \int_0^{\tau(v_0)} (3x(t)^2 - 2kx(t) + 1)\, dt \right] > 0.$$

Thus P is strictly increasing. A periodic orbit yields a fixed point of P: an increasing map of an interval has no periodic points of higher period. Conversely, a fixed point of P yields a periodic orbit.

For $v_0 > 0$ sufficiently small, the outward function L gives $P(v_0) > v_0$. Indeed, on $x = a$ the original displacement is $\xi = (0, v)^T$, so $L = M_{22} v^2$. If the return occurs before leaving $\mathcal{U}$, L has increased; if it occurs after leaving, the trajectory remains outside $\mathcal{U}$ and the returned velocity is again larger than v_0. Thus $P(v) - v$ is not identically zero and has no zeros near $v = 0$.

Every fixed point corresponds to an orbit contained in the same absorbing ellipse, so its velocity on Σ has a common finite upper bound. All fixed points consequently lie in a compact subinterval of $(0, \infty)$. A nonzero real analytic function has only finitely many zeros on such an interval. Hence there are finitely many periodic orbits, all isolated; they are limit cycles.

Finally, $P(v) < v$ for v sufficiently large. Otherwise, monotonicity would keep all iterates $P^n(v)$ at least as large as this initial value, preventing the trajectory from entering the absorbing ellipse. Since $P(v) - v$ is positive near zero and negative for large v, at least

one of its isolated zeros has positive sign on its left and negative sign on its right. The corresponding cycle is orbitally asymptotically stable. This proves part 3 of the theorem.

Global uniqueness for $a \geq a_0$. Fix $k > k_H$ and put $f_k = F_k'$. Since $F_k(p)/p$ is a monic quadratic with negative value at $p = 0$, it has two simple roots of opposite signs. Write

$$F_k(p) = p(p+b)(p-r), \qquad b > 0, \quad r > 0, \quad r - b = k - 3a, \quad br = 2a(k - k_H).$$

More explicitly, with $\Delta_k = k^2 + 2ak - 3a^2 - 4$,

$$b = \frac{3a - k + \sqrt{\Delta_k}}{2}, \qquad r = \frac{k - 3a + \sqrt{\Delta_k}}{2}.$$

Thus F_k is negative on $(-\infty, -b)$ and $(0, r)$, and positive on $(-b, 0)$ and (r, ∞). Moreover, $b < 2a$, because $F_k(-2a)/(-2a) = a^2 + 1 > 0$ while $F_k(p)/p < 0$ at $p = 0$. Rolle's theorem gives exactly two zeros ρ_L, ρ_R of f_k, with

$$-b < \rho_L < 0 < \rho_R < r, \qquad f_k(p) = 3(p - \rho_L)(p - \rho_R). \tag{40}$$

In particular, $f_k > 0$ on both outer intervals.

We use a uniqueness criterion based on the potential coordinate. This form of the branch comparison argument goes back to Zeng; we apply [6] with $\phi(q) = q$. Its relevant form is the following. Let $p_R(h) > 0 > p_L(h)$ satisfy $G(p_R(h)) = G(p_L(h)) = h$, and define

$$\mathfrak{F}_1(h) = F_k(p_R(h)), \qquad \mathfrak{F}_2(h) = F_k(p_L(h)).$$

The system has at most one limit cycle, which is hyperbolic and attracting if it exists, provided that, for some $c > 0$,

(1) $\mathfrak{F}_1(h) \leq 0 \leq \mathfrak{F}_2(h)$ for $0 < h < c$, the two functions differ near $h = 0$, and $\mathfrak{F}_1(h) > 0$ for $h > c$;
(2) $\mathfrak{F}_2'(h) \leq 0$ wherever $\mathfrak{F}_2(h) < 0$;
(3) $\mathfrak{F}_1(h)\mathfrak{F}_1'(h)$ is nondecreasing for $h > c$;
(4) $\mathfrak{F}_1'(h) \geq \mathfrak{F}_2'(u)$ whenever $\mathfrak{F}_1(h) = \mathfrak{F}_2(u)$ and $u \geq h > c$.

The hypotheses $pg(p) > 0$, $G(\pm\infty) = \infty$, and the required regularity already hold. The cited criterion estimates the characteristic exponent of every periodic orbit; its strict negativity gives hyperbolicity as well as attraction. We now verify the four conditions, allowing the reflection $(p, q) \mapsto (-p, -q)$ when necessary.

First, we establish the monotonicity on both outer branches. Define

$$\mathcal{P}_k(p) = \frac{F_k(p) f_k(p)}{g(p)} = \frac{(p+b)(p-r) f_k(p)}{p^2 + 3ap + 3a^2}.$$

The denominator is strictly positive. For $p > r$, logarithmic differentiation using (40) gives

$$\frac{\mathcal{P}_k'(p)}{\mathcal{P}_k(p)} = \frac{1}{p+b} + \frac{1}{p-r} + \frac{1}{p-\rho_L} + \frac{1}{p-\rho_R} - \frac{2p+3a}{p^2+3ap+3a^2} > 0.$$

Indeed, $1/(p-r) > 1/p$, $1/(p-\rho_R) > 1/p$, and $(2p+3a)/(p^2+3ap+3a^2) < 2/p$; the other two terms are positive. Hence $\mathcal{P}_k$ is strictly increasing on the right outer branch for every $a > 0$.

For the left branch write $z = -p > b$ and $\beta = -\rho_L \in (0, b)$. Then

$$\mathcal{P}_k(-z) = \frac{3(z-b)(z+r)(z-\beta)(z+\rho_R)}{z^2 - 3az + 3a^2} > 0.$$

When $b < z \leq 2a$, its logarithmic derivative is positive: the terms $1/(z-b)$ and $1/(z-\beta)$ are each larger than $1/z$, while

$$\frac{2z - 3a}{z^2 - 3az + 3a^2} \leq \frac{2}{z} \qquad \text{for } 0 < z \leq 2a.$$

The remaining two terms are positive. This proves $\mathcal{P}_k'(p) < 0$ for $-2a \leq p < -b$.

To handle the rest of the left branch, return temporarily to $x = p + a$. Set

$$\begin{aligned} H(x) &= x^2 + ax + a^2, \\ Q_k(x) &= x^2 + (a-k)x + a^2 - ka + 1, \\ f_0(x) &= 3x^2 - 2kx + 1, \qquad g_0(x) = x^3 - a^3, \\ \Pi_k(x) &= \mathcal{P}_k(x-a) = \frac{Q_k(x) f_0(x)}{H(x)}. \end{aligned}$$

Define $D_k(x) = H(x)^2 \Pi_k'(x)$. If $x = -a - w$, $w \geq 0$, and $s = k - k_H \geq 0$, direct differentiation and collection of powers give the exact identity

$$D_{k_H+s}(-a-w) = -\frac{B_a(w)}{4a^2} - sC_a(w) - 2a^2(a+2w)s^2, \tag{41}$$

where

$$\begin{aligned} B_a(w) = {} & 24a^2w^5 + (102a^3 + 10a)w^4 + (180a^4 + 20a^2)w^3 \\ & + (228a^5 + 6a^3 - 6a)w^2 + (204a^6 + 32a^4 - 12a^2)w \\ & + 72a^7 + 40a^5, \\ C_a(w) = {} & 5w^4 + 10aw^3 + (18a^2 - 3)w^2 + 28a^3w + 11a^4 + 3a^2. \end{aligned}$$

For $a^2 \geq 3/17$, every coefficient in these two polynomials is nonnegative, and their constant terms are positive. For the potentially ambiguous coefficients, observe that

$$\begin{aligned} 18a^2 - 3 &> 0, \\ 228a^5 + 6a^3 - 6a &= 6a(38a^4 + a^2 - 1) > 0, \\ 204a^6 + 32a^4 - 12a^2 &= 4a^2(17a^2 - 3)(3a^2 + 1) \geq 0. \end{aligned}$$

Here $38a^4 + a^2 - 1 \geq 104/289 > 0$ at and above $a^2 = 3/17$. Thus (41) is strictly negative. It follows that $\Pi_k'(x) < 0$ for $x \leq -a$, or equivalently $\mathcal{P}_k'(p) < 0$ for $p \leq -2a$. Combining the two portions of the left branch, we have proved

$$\mathcal{P}_k'(p) < 0 \quad (p < -b), \qquad \mathcal{P}_k'(p) > 0 \quad (p > r), \qquad a \geq a_0. \tag{42}$$

Next, we verify the slope comparison in condition 4. This part of the argument holds for every $a > 0$. Suppose $t < a < x$ and $F_k(x-a) = F_k(t-a)$. Put

$$S = x + t, \qquad d = x - t > 0, \qquad B = \frac{S(x^2 + t^2)}{4}.$$

Using $F_k(x-a) = \Phi_k(x) - \Phi_k(a)$, equality of the two values gives

$$x^2 + xt + t^2 + 1 = kS. \tag{43}$$

Its left side is positive, so $S > 0$. The potential $\widehat{G}(x) = x^4/4 - a^3x + 3a^4/4$ satisfies

$$\widehat{G}(x) - \widehat{G}(t) = d(B - a^3).$$

Furthermore, substitution of (43) gives

$$\begin{aligned} & f_0(x)g_0(t) - f_0(t)g_0(x) \\ & \qquad = -d\left[\frac{d^2(d^2+2)}{4} + (a^3 - B)(3S - 2k)\right]. \end{aligned} \tag{44}$$

Also $f_0(x) - f_0(t) = d(3S - 2k)$.

If x is on the right outer branch and $\widehat{G}(t) \geq \widehat{G}(x)$, then $f_0(x) > 0$ and $a^3 - B \geq 0$. When $f_0(t) \geq 0$, the signs of g_0 already give $f_0(x)/g_0(x) > f_0(t)/g_0(t)$. When $f_0(t) < 0$,

we have $3S - 2k > 0$, so the right side of (44) is strictly negative. Since $g_0(x)g_0(t) < 0$, the same conclusion follows:

$$\frac{f_0(x)}{g_0(x)} > \frac{f_0(t)}{g_0(t)}. \tag{45}$$

This is condition 4 when branch 1 is the right branch, because differentiation with respect to the potential gives $\mathfrak{F}_i' = f_k/g$.

If branch 1 is obtained by reflection from the left outer branch, then $f_0(t) > 0$ and the relevant potential ordering is $\widehat{G}(x) \geq \widehat{G}(t)$. For $f_0(x) \geq 0$, inequality (45) again follows from the signs. Otherwise, both $a^3 - B \leq 0$ and $3S - 2k < 0$. Their product is nonnegative, and (44) again gives (45). Under reflection the derivatives of the two branches are $-f_0(t)/g_0(t)$ and $-f_0(x)/g_0(x)$, respectively, so condition 4 holds in this case as well.

We can now choose the orientation of the criterion. Let $h_R = G(r)$ and $h_L = G(-b)$. If $h_R \leq h_L$, use the original coordinates and take $c = h_R$. If $h_L < h_R$, use

$$\widetilde{p} = -p, \quad \widetilde{q} = -q, \qquad \widetilde{F}(\widetilde{p}) = -F_k(-\widetilde{p}), \quad \widetilde{g}(\widetilde{p}) = -g(-\widetilde{p}),$$

and take $c = h_L$. In either orientation, branch 1 corresponds to the nonzero root with the smaller potential. The three-zero sign pattern proves condition 1, including strictly different signs near $h = 0$. On the part of branch 2 where its value is negative, $f_k > 0$ and its restoring function is negative; hence condition 2 holds. For condition 3, note that

$$\mathfrak{F}_1(h)\mathfrak{F}_1'(h) = \frac{F_k(p)f_k(p)}{g(p)} = \mathcal{P}_k(p)$$

on the chosen outer branch, with the corresponding reflected formula when needed. Its derivative with respect to h is positive by (42). Finally, (45) proves condition 4 in both orientations.

All hypotheses of the global uniqueness criterion are therefore verified for $a \geq a_0$ and every $k > k_H$. Existence was proved above, so there is exactly one hyperbolic attracting limit cycle in the entire plane. Since every non-equilibrium trajectory approaches a periodic orbit, it must approach this unique cycle. Together with the nonexistence result for $k \leq k_H$, this proves part 4.

The supercritical Hopf calculation gives a neighborhood $\mathcal{V}$ of the origin in the centered coordinates and $\delta_0(a) > 0$ such that exactly one attracting hyperbolic periodic orbit lies in $\mathcal{V}$ when $k_H < k < k_H + \delta_0(a)$. To obtain global uniqueness, we must also exclude periodic orbits outside $\mathcal{V}$.

The uniform boundedness estimate supplies a fixed compact set $\mathcal{K}$ containing every periodic orbit for k in a neighborhood of k_H. At $k = k_H$ the origin is globally asymptotically stable. Attraction is therefore uniform on the compact set $\mathcal{K}$: Lyapunov stability supplies a small neighborhood from which trajectories stay near the origin, and global attraction and a finite covering of $\mathcal{K}$ give a common entry time. Choose an open neighborhood $\mathcal{V}_0$ whose closure lies in $\mathcal{V}$. There is a finite time T such that $\varphi_{k_H}(T, \mathcal{K}) \subset \mathcal{V}_0$, where φ_k denotes the centered flow. Continuous dependence on k and on initial conditions, uniformly over $\mathcal{K}$, gives $\delta_1(a) > 0$ with

$$\varphi_k(T, \mathcal{K}) \subset \mathcal{V} \qquad \text{whenever } |k - k_H| < \delta_1(a).$$

For any periodic orbit Γ in this parameter range,

$$\Gamma = \varphi_k(T, \Gamma) \subset \varphi_k(T, \mathcal{K}) \subset \mathcal{V}.$$

Every periodic orbit thus lies in the Hopf neighborhood. Taking

$$\delta(a) = \min\left\{\delta_0(a), \delta_1(a), \frac{K(a) - k_H}{2}\right\} > 0$$

proves that the Hopf cycle is the unique limit cycle in the entire plane for $k_H < k < k_H + \delta(a)$.

Assume $k > K(a)$ and retain the factorization (5). The inequality $k > a + 1/a$ gives $R_k(-a) = a^2 - ka + 1 < 0$, where $R_k = F_k/p$; hence $b > a$. Also,

$$r > k - 3a > 3\sqrt{3}.$$

The derivative F_k' has one zero between $-b$ and 0 and one between 0 and r. Let the latter be m. Then

$$m = \frac{r - b + \sqrt{r^2 + br + b^2}}{3} < \frac{2r}{3},$$

and F_k is strictly increasing for $p \le -b$ and for $p \ge m$.

The potentials at the nonzero roots satisfy

$$G(r) - G(-b) = \frac{r+b}{4}\big[(k-a)(k^2 - a^2 - 2) - 4a^3\big] > 0.$$

The last inequality follows from $k > 3a + 3\sqrt{3}$: $k - a > 2a$ and $k^2 - a^2 - 2 > 8a^2 + 25$. Let $\eta \in (0, r)$ be the unique point for which $G(\eta) = G(-b)$.

We apply the asymmetric uniqueness criterion in [3] with the coefficient $\lambda = 1$. In the present notation, if $\eta \ge m$, the outer monotonicity already gives at most one cycle. If $\eta < m$, a sufficient condition is

$$\max_{\eta \le p \le m} \left[\frac{1}{-F_k(p)} \int_\eta^p \frac{g(s)}{-F_k(s)}\, ds \right]^{1/2} < 1. \tag{46}$$

Under either condition the periodic orbit, if it exists, is hyperbolic and attracting.

It remains to verify (46) when $\eta < m$. For $0 < s \le m < 2r/3$,

$$-F_k(s) = s(s+b)(r-s) \ge \frac{r}{3} s(s+b).$$

Since $b \ge a$,

$$\frac{s^2 + 3as + 3a^2}{s+b} \le s + 3a.$$

Therefore, for $\eta \le p \le m$,

$$\begin{aligned} \frac{1}{-F_k(p)} \int_\eta^p \frac{g(s)}{-F_k(s)}\, ds &\le \frac{9}{r^2} \frac{p^2/2 + 3ap}{p(p+b)} \\ &= \frac{9}{r^2} \frac{p/2 + 3a}{p+b} \le \frac{27}{r^2} < 1. \end{aligned}$$

This verifies the uniqueness criterion. Existence has already been proved for every $k > k_H$, so there is exactly one hyperbolic attracting limit cycle whenever $k > K(a)$. Part 3 shows that every non-equilibrium trajectory approaches it. Together with the near-Hopf argument, this completes the proof of Theorem 5.1. □

For $I \ge I_0$, our result is a complete global cycle count for every $k > 0$. Below and at Hopf every trajectory converges to E_*; above Hopf every non-equilibrium trajectory approaches the unique attracting cycle. Hyperbolicity and uniqueness also imply that the local supercritical Hopf branch continues throughout $k > k_H$, with no fold of periodic orbits in this parameter range. This is a global uniqueness conclusion, not merely the local uniqueness provided by the Hopf theorem.

For $-1 < I < I_0$, existence is still completely determined: there is no periodic orbit for $k \le k_H$, and at least one limit cycle for every $k > k_H$. The global count is exactly one near Hopf and for $k > K(a)$, as stated in part 4. An exact count in the intervening range is not established by the present arguments. The sufficient condition $a \ge a_0$ cannot simply

be removed from the proof above. For example, at $a = 1/5$ and $k = 3 > k_H = 14/5$, the point $x = -1/2$ lies on the left outer branch, but

$$\Pi_3'(-1/2) = 26 > 0,$$

so the required monotonicity fails there. This failure does not prove the existence of additional cycles; it identifies the step requiring a different argument before the exact count can be asserted for every $I > -1$.

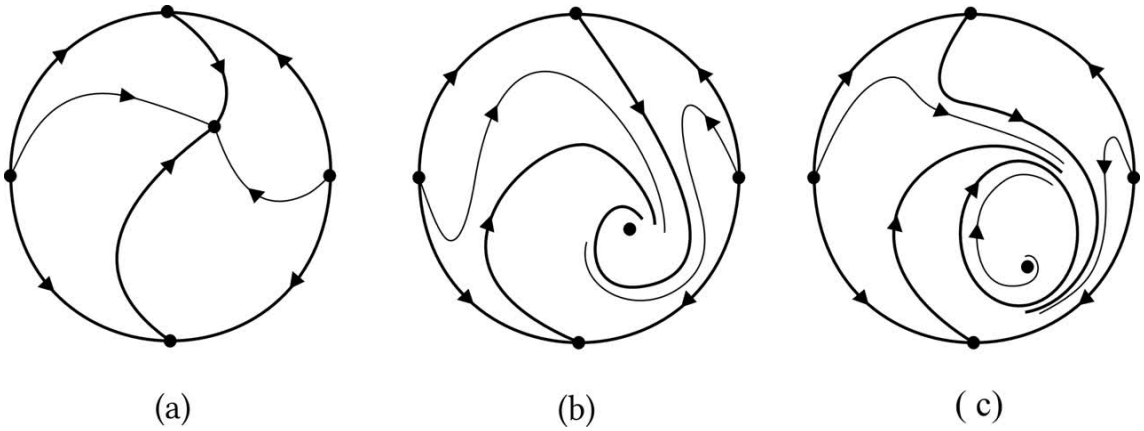


FIGURE 1.

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